



PowerGrid AI

Predictive Grid Outage & Equipment Failure Advisor

Predict. Prevent. Keep the Grid On.

Hackathon Identity

Track

AI Track — Bob AI Hackathon

Team & Lead

PowerGrid AI | Ishika

Target Domain

High-Voltage Electrical Power Grids — **836+ Gujarat Assets**

"Transforming utility operations from reactive emergency repairs to proactive AI-driven failure prevention."

Innovation Badges

22 ML Features

4 domain-engineered IEC
60599 Rogers Ratios

Auto-Retraining

Weekly XGBoost retraining via
APScheduler (Sunday 00:00
UTC)

MLflow Tracked

Full experiment lineage, model
versioning, metric logging

9 REST Endpoints

Full FastAPI microservices
with Swagger docs

Weather-Compound Risk

Wind speed & rainfall as ML
features — unique
differentiator

Physical Safety Overrides

Domain-calibrated post-
inference thresholds for
critical faults



Reactive Grid Management Is Failing

High-voltage power transformers fail silently. Dissolved Gas Analysis (DGA) chemistry anomalies accumulate invisibly until catastrophic breakdown —

and today's operators are flying blind across fragmented SCADA tools.

Silent Transformer Failure

DGA gas anomalies go undetected until breakdown. **Acetylene (C₂H₂)** indicates arcing | **Ethylene (C₂H₄)** indicates overheating | **Methane (CH₄)** indicates degradation. Each unplanned failure costs **\$2.4M per transformer replacement** plus regional blackout impact.

Who Bears the Cost

Grid Dispatch Engineers and SCADA Operators at GETCO, PGCIL, and regional utility boards. Industrial and residential consumers across Gujarat facing sudden outages affecting **millions of lives**.

Why It Matters Now

Rapid integration of renewable energy (solar/wind microgrids) adds dynamic thermal and electrical strain on aging assets. India's grid expansion demands AI-first predictive infrastructure — not reactive firefighting.

The Operational Pain Point

3+

Hours Per Incident

Manually correlating raw sensor logs across 12+ disconnected SCADA tools

12+

Disconnected Tools

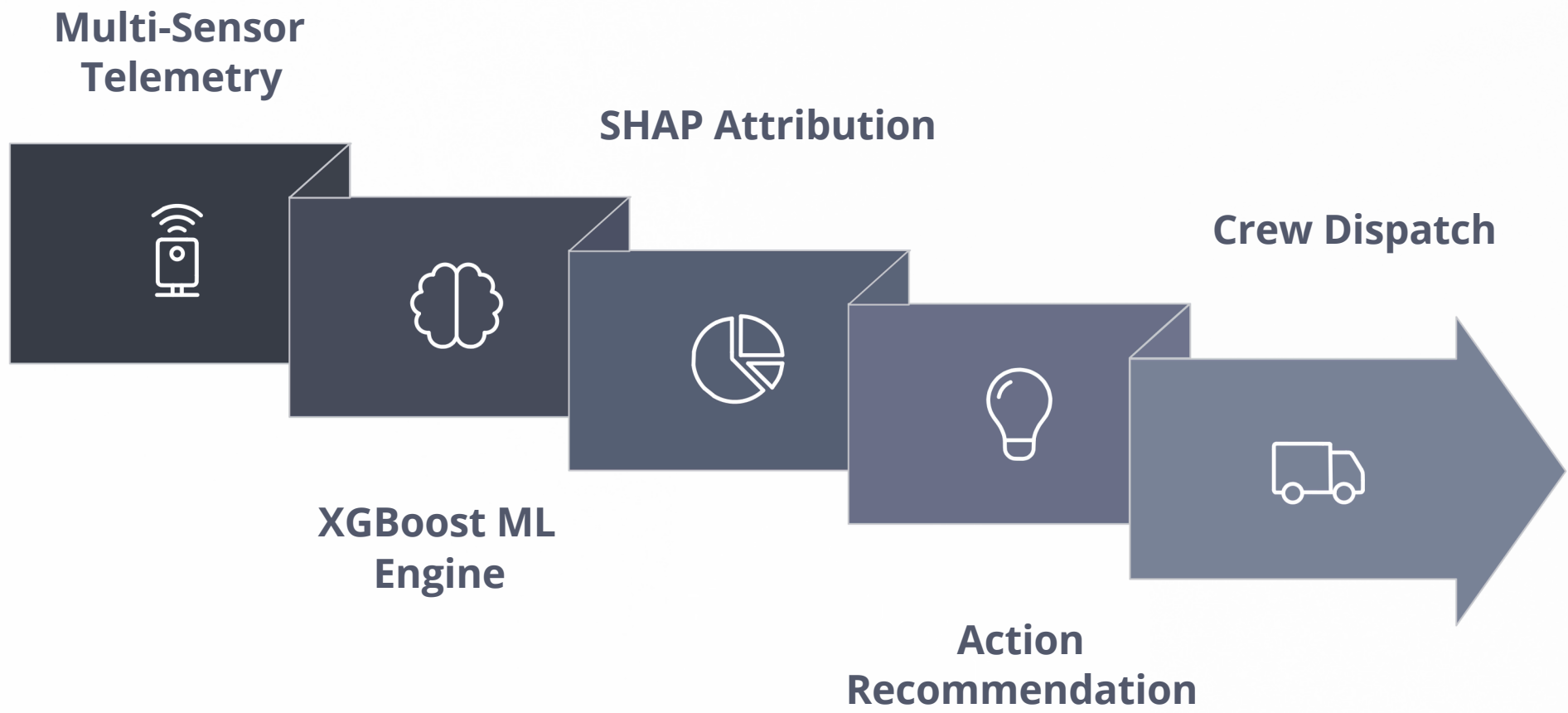
Engineers toggle between siloed SCADA systems, spreadsheets, and IEC reference charts

\$2.4M

Unplanned replacement cost per failed unit — before counting outage damages

PowerGrid AI Grid Advisor

An AI-powered decision-support platform that ingests multi-sensor grid telemetry, diagnoses transformer DGA gas chemistry using machine learning, visualizes statewide grid health, and automates field crew dispatch **before outages occur**.



Core Value Propositions

-  **96.25% Fault Classification**
XGBoost accuracy across 6 critical fault modes — from Normal to Arcing/Partial Discharge
-  **Statewide Gujarat GIS Map**
All 33 districts with color-coded asset health markers and animated transmission lines
-  **Interactive AI Testing Studio**
Real-time parameter simulation with SHAP explainability waterfall — live DGA chemistry inspection

Gujarat Grid Coverage

836+ Assets
Mapped with real geo-coordinates across all 33 districts

Color-Coded Markers
Green (Healthy) | Yellow (Warning) | Red (Critical)

4 Asset Types
Transformers · Substations · Feeders · Transmission Lines



Technical Architecture & API Surface

System Architecture Layers

1

Next.js 16 Web Dashboard

App Router · Tailwind CSS · Leaflet GIS · Recharts — Multi-tab operational dashboard and AI studio controls

2

Python FastAPI Backend

9 REST Endpoints — Asset Manager · Risk Engine · Crew Dispatcher · Weather · Telemetry Ingestion

3

Data & ML Layer

Neon Serverless PostgreSQL (8 tables, SQLAlchemy ORM) | XGBoost ML Artifacts (DGA & Failure Models, MLflow tracked)

9 REST API Endpoints

/api/assets

CRUD for 836+ grid assets with geo-coordinates and district mapping

/api/sensors

Real-time telemetry ingestion (temperature, vibration, load percent)

/api/weather

30-day district weather history plus 7-day predictive forecast

/api/predictions

Live XGBoost inference + SHAP drivers + action recommendations

/api/risks

Composite risk scoring (gas + thermal + weather + impact)

/api/crews

Field team dispatch with zone-based skill-level matching

/api/incidents · /api/dashboard · /api/maintenance

Historical records, KPI aggregation, technician logs & due-date alerts

Database Schema — 8 Relational Tables

Table	Key Columns	Relationship
ASSETS	asset_id, name, district, asset_type, geo_coordinates	Parent — has many of all below
SENSOR_READINGS	temperature, oil_temp, vibration, load, voltage, current	Many → ASSETS
PREDICTIONS	failure_probability, health_score, DGA values, fault_type	Many → ASSETS
INCIDENTS	fault_type, severity, customers_affected, root_cause, duration	Many → ASSETS
RISK_SCORES	failure_prob, impact_score, weather_risk, overall_risk, level	Many → ASSETS
RECOMMENDATIONS	action_type, urgency, priority, status	Many → ASSETS
WEATHER_READINGS	avg_temp, wind_speed, rainfall, air_pressure, season	Districts (independent)
CREWS	crew_id, name, zone, available, skill_level	Assigned to Zones



XGBoost DGA Fault Classification Engine

Model Architecture

Algorithm

XGBoost — Gradient-Boosted Decision Trees with MLflow experiment tracking and APScheduler auto-retraining (Sunday 00:00 UTC)

Model 1 — Failure Probability Regressor

reg:squarederror | 150 trees | max depth 5 | learning rate 0.05

Model 2 — Fault Mode Classifier

multi:softprob | 120 trees | max depth 4 | learning rate 0.08 | 6 output classes

Training Split

80/20 train-test | random_state=42 for reproducibility | **96.25% test accuracy**

Feature Count

22 features: 10 DGA gas readings + 6 operational metrics + 4 domain-engineered ratios + 2 weather features

6-Class Fault Mode Classification

Class	Fault Mode	Trigger Threshold
0	Normal / Low Risk	All gases within IEC 60599 safe thresholds
1	Insulation Degradation	Failure prob > 0.40, moderate gas levels
2	Thermal Overheating	C ₂ H ₄ /C ₂ H ₆ > 3.0 OR thermal stress index > 0.75
3	Dielectric Breakdown	Dielectric rigidity < 35 kV OR water > 35 ppm
4	Mechanical / Vibration Stress	Vibration > 5.0 mm/s
5	Arcing / Partial Discharge	C ₂ H ₂ > 12 ppm OR C ₂ H ₂ /C ₂ H ₄ > 1.0 with high risk

Domain-Engineered Features (IEC 60599 Rogers Ratios)

CH₄ ÷ H₂

Methane/Hydrogen — Rogers Ratio, thermal fault severity indicator

C₂H₄ ÷ C₂H₆

Ethylene/Ethane — hot-spot severity (>3.0 = confirmed overheating)

C₂H₂ ÷ C₂H₄

Acetylene/Ethylene — arcing vs thermal discriminator (>1.0 = arcing confirmed)

(OilTemp – 45°C) ÷ 45

Thermal Stress Index — normalized loading beyond safe baseline

Wind Speed + Rainfall

Weather-compound features — wind >60 km/h compounds mechanical stress; rain >50 mm compounds dielectric breakdown during monsoon

SHAP Root-Cause Example — Asset GJ-TRF-0234

Prediction: Arcing / Partial Discharge | Failure Probability: **0.94** | Risk Window: 6 hours

Feature	Value	SHAP Impact
Acetylene (C ₂ H ₂)	28.5 ppm	+0.2413
Ethylene/Ethane ratio	4.21	+0.1847
Oil Temperature	87.3°C	+0.1192
Wind Speed	72.0 km/h	+0.0834
Load Percent	108.0%	+0.0621



SHAP ensures operators understand **WHY** the AI recommends action — critical for grid safety certification and regulatory compliance.



Station-Level DGA Inspector & AI Studio

The AI Studio transforms a 3-hour manual investigation into a sub-15-second guided diagnostic—from raw gas chemistry to crew dispatch, with full SHAP explainability at every step.

Interactive Demo Workflow

- 1

Search & Select Asset

Query transformer from statewide inventory by ID, district, or asset type
- 2

Inspect Live Chemistry

C₂H₂ · C₂H₄ · CH₄ · H₂ · Oil Temperature · Winding Vibration — all DGA gases with IEC 60599 color badges
- 3

XGBoost + SHAP Inference

Fault class predicted; SHAP waterfall shows which gas ratios triggered the diagnosis
- 4

One-Click Crew Dispatch

Zone field crew assigned with skill-level match (e.g., HV Arcing Specialist)

Before vs. After: The ThreatOps Transformation

Workflow Step	Before PowerGrid	After PowerGrid
Data Collection	Login to 12+ separate SCADA tools	Single unified dashboard, auto-ingested
DGA Analysis	Manually read ppm tables, cross-reference IEC charts	AI auto-classifies with color-coded IEC 60599 badges
Root Cause	Guesswork based on operator experience	SHAP waterfall shows exact gas ratios driving the fault
Risk Assessment	Subjective verbal estimate (high/medium/low)	Quantified probability 0.00–0.99 with calibrated risk window
Crew Dispatch	Phone calls, manual schedule checking	One-click dispatch with zone-based skill match
Total Time	3+ hours per incident	Under 15 seconds end-to-end

Key Capabilities

- ### IEC 60599 Standards Badges

Automated color-coded risk status — Green (Normal) · Yellow (Warning) · Red (Critical) — per gas level, per reading
- ### District Weather Threat Radar

Integrated 30-day historical weather trend plus 7-day predictive forecast cards for compound risk assessment



Intelligent Risk Calibration & Physical Safety Overrides

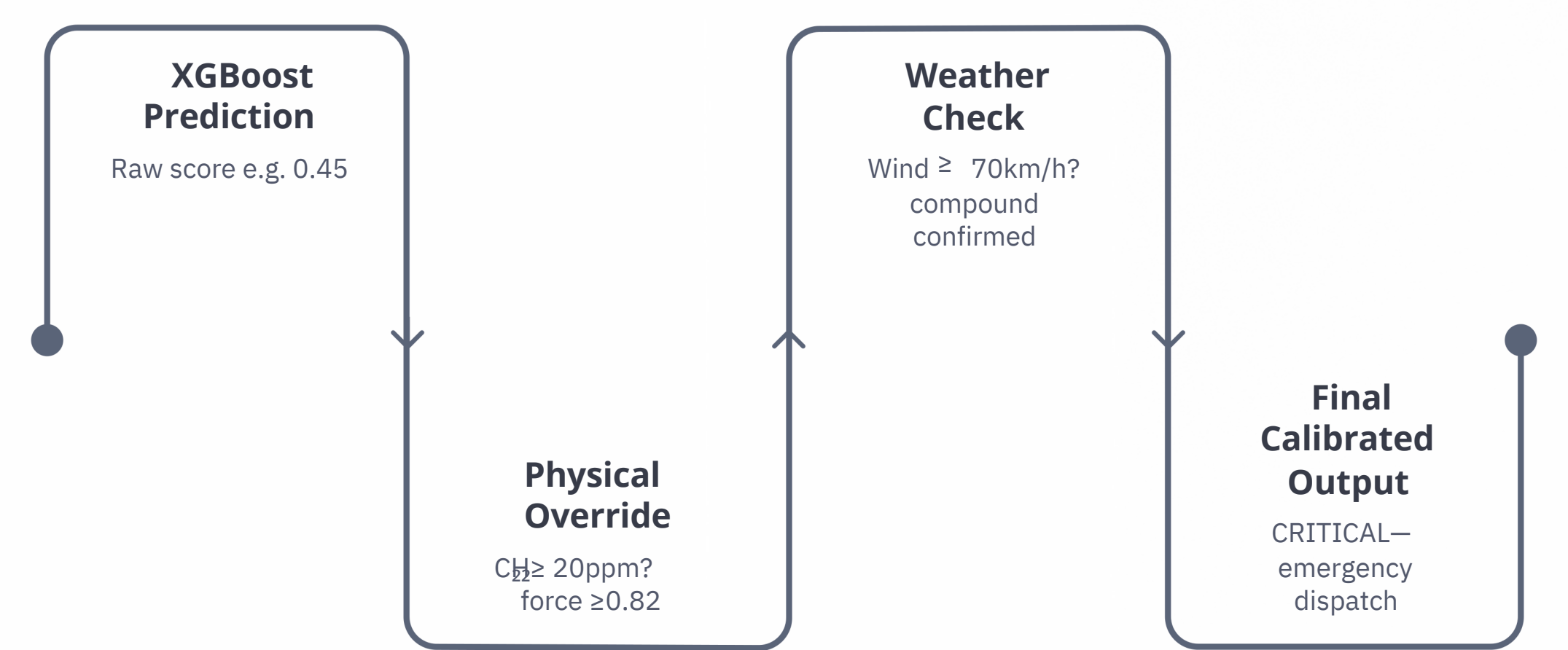
ThreatOpsoperates a dual-layersafety architecture: XGBoost MLinference **plus** domain-calibrated physicaloverriderules.TheAI**never** under-predicts catastrophic failures — even when the model's raw output is conservative. This is production-grade safety engineering.

4-Tier Risk Escalation Matrix

LOW Probability < 0.35 Risk Window: 7 days <i>"Normal: Equipment within safe parameters"</i>	MEDIUM Probability 0.35–0.60 Risk Window: 48 hours <i>"Advisory: Inspect at next maintenance cycle"</i>
HIGH Probability 0.60–0.80 Risk Window: 24 hours <i>"Warning: Schedule diagnostic inspection"</i>	CRITICAL Probability ≥ 0.80 Risk Window: 6 hours <i>"Emergency: Immediate dispatch, switch to redundant feeder"</i>

Physical Safety Override Rules — 6 Domain-Calibrated Thresholds

Condition	Threshold	Override Action	Risk Level	IEC / Engineering Basis
Acetylene (C ₂ H ₂)	≥ 20 ppm	Force min probability → 0.82+	CRITICAL	IEC 60599: confirmed internal arcing
Acetylene (C ₂ H ₂)	≥ 10 ppm	Force min probability → 0.76+	HIGH	Early arcing indicator — cannot be ignored
Oil Temperature	≥ 90°C	Force min probability → 0.84+	CRITICAL	Oil decomposition + insulation carbonization zone
Load Percent	≥ 115%	Force min probability → 0.84+	CRITICAL	Exceeds nameplate rating — flashover risk
Vibration	≥ 5.5 mm/s	Force min probability → 0.78+	HIGH	Core/winding displacement precursor to failure
Wind ≥ 70 km/h OR Rain ≥ 80 mm	Severe weather	Force min probability → 0.62+	MEDIUM+	Gujarat monsoon/cyclone compounds insulation stress



☒ This dual-layer approach — ML inference **plus** domain safety rules — is what separates ThreatOps from academic demos. It is deployment-ready for real grid operations.

IBM Technology Integration

ThreatOpsnativelyintegratesIBM's enterpriseAI andobservability stacktopowerintelligent grid management — combining LLM-based reasoning,

multi-sensor anomaly classification, and real-time SCADA telemetry ingestion.



IBM Bob AI Integration

Conversational Assistant & CLI

Provides natural language incident summaries for complex DGA gas anomalies. Enables one-click runbook execution for emergency feeder switching. Operators can ask *"Summarize the risk for Ahmedabad substation TRF-0234"* and receive structured operational briefings — no dashboard navigation required.



IBM watsonx.ai

Granite 3.0 Model Engine

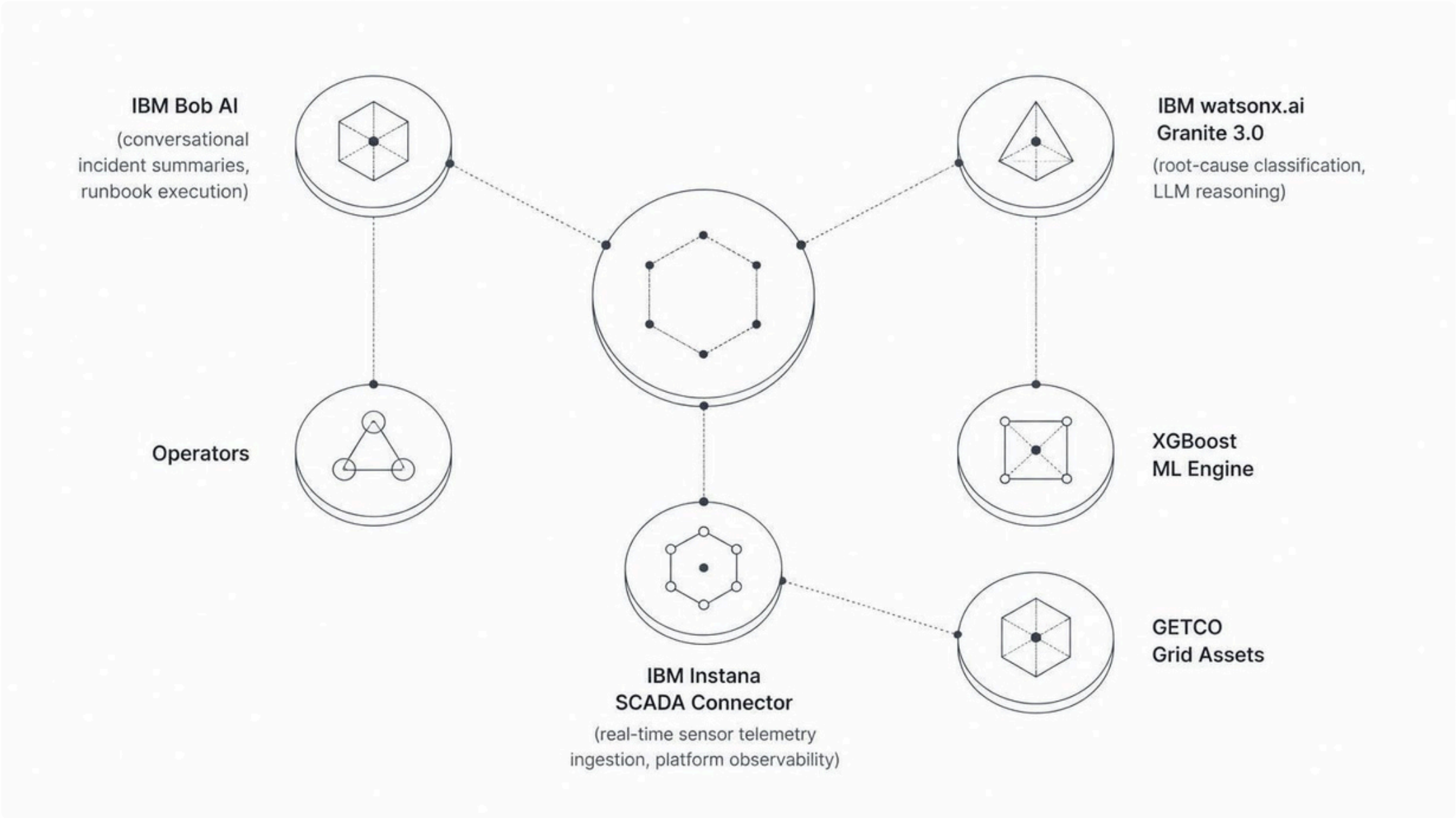
Powers root-cause classification across multi-sensor telemetry streams. Evaluates thermal, electrical, and gas ratios to output recommended operator actions. Complements XGBoost with LLM-based contextual reasoning for edge cases where gas ratios alone are ambiguous.



IBM Instana SCADA Connector

Unified Telemetry Ingestion

Ingests real-time sensor telemetry (temperatures, gas ppm, load percent, vibration) from high-voltage GETCO substations. Provides infrastructure-level observability for the ThreatOps platform itself — API latency, error rates, and throughput dashboards.



LIVE DATA FLOW



Sensor to Crew Dispatch in <15 Seconds

Real-world end-to-end trace for Asset GJ-TRF-0234 at Ahmedabad Substation

Input Readings — SCADA Sensor Push

Sensor Category	Parameter	Value
DGA Gas	Acetylene (C ₂ H) ₂	28.5 ppm
DGA Gas	Ethylene (C ₂ H) ₄	85 ppm
DGA Gas	Methane (CH) ₄	62 ppm
Operational	Oil Temperature	87.3°C
Operational	Vibration	3.2 mm/s
Weather	Wind Speed	72 km/h
Weather	Rainfall	45 mm

Ingested via IBM Instana SCADA Connector → POST /api/sensors → Neon Postgres

Quantitative Impact & Why ThreatOps Wins

96.25%

XGBoost test accuracy across 6 fault modes

836+

All 33 Gujarat districts mapped with real geo-coordinates

75%

Downtime cut through proactive AI-driven dispatch

\$10M+

Cost avoidance from prevented transformer failures

Business & Operational Impact — Before vs. After

Metric	Before	After ThreatOps	Benefit
Incident Diagnosis	3+ hours manual	<15 seconds AI inference	95% faster MTTR
Fault Classification	Operator experience (subjective)	96.25% XGBoost (6 fault modes)	Eliminates human error
Grid Coverage	Sampled manual testing	836 assets, 33 districts	100% statewide visibility
Explainability	None — black box decisions	SHAP root-cause attribution per prediction	Operator confidence + compliance
Crew Dispatch	Reactive — post-outage phone calls	Proactive pre-positioning with skill match	Zero emergency delay
Cost Avoidance	\$2.4M per failed transformer	Preventative AI maintenance	\$10M+ saved annually
Weather Risk	Separate system, uncorrelated	Compound ML feature (wind + rain)	Monsoon/cyclone preparedness

What Makes ThreatOps Defensible

IEC 60599 Rogers Ratios

4 domain-engineered features grounded in international transformer chemistry standards — not generic ML

Dual-Layer Safety

Physical override rules ensure the AI never under-predicts catastrophic arcing or thermal runaway — deployment-ready, not academic

SHAP Explainability

Every prediction auditable — operators see exactly which gas ratio triggered the fault, enabling regulatory compliance

Full Production Stack

MLflow versioning · APScheduler retraining · 9 REST APIs · 8 relational tables · GIS statewide coverage — not a prototype

"ThreatOps is not an academic demo. It is a deployment-ready AI infrastructure platform built for the realities of Gujarat's high-voltage grid — with physical safety overrides, SHAP transparency, and full IBM enterprise integration."

Our Team

Ishika [Lead] — AI/ML

- Failure Risk Prediction
- Anomaly Detection

Dhriti — Full-Stack

- Dashboard Development
- System Integration

Manthan — Grid Intelligence

- Fault Diagnosis
- DGA & Thermal Aging

Abhishek — Optimization

- Grid Impact & CPI
- Crew Pre-Positioning